Unit 4, Lesson 12: Operations with Absolute Value in the Real World

Benchmark

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute values.

Learning Target

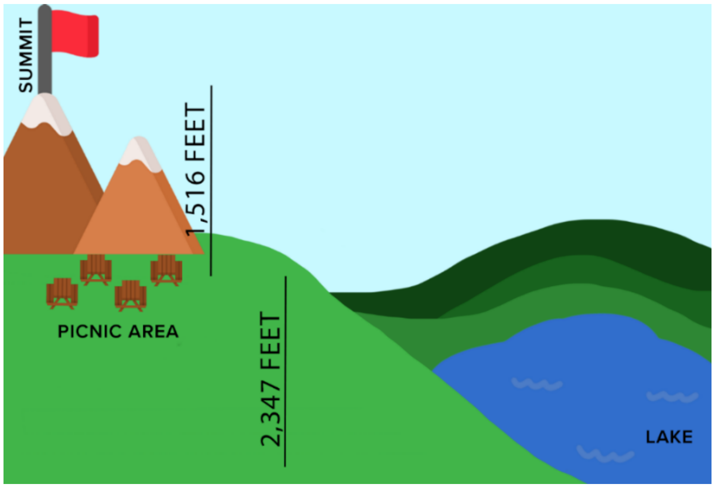
- ☐ I can perform operations that involve the absolute value of two numbers.

4.12.1 Warm-Up: Math Fail

1. What do you notice about the sign?

4.12.2 Guided Instruction: Operations with Absolute Value in the Real World

1. Javier and his family drove to a picnic area on a mountain. From the picnic area, the family hiked to the mountain's summit, which was an elevation change of $1{,}516$ feet (ft). They hiked back to the picnic area to eat lunch. After lunch, they hiked down to the lake, which was an elevation change of $2{,}347$ ft.



a. In the table, write the value that represents the position of each location if the value of the position of the picnic area is 0. Then, plot each value at the estimated location on the number line.

Location	Value
Picnic area	0
Mountain summit	
Lake	

b. Represent the distance of each location from the picnic area.

Location	Distance from Picnic Area
Mountain summit	
Lake	

c. What operation would you use with the representations in the table to find the distance from the lake to the mountain summit?

d. What is the distance from the lake to the mountain summit?

e. What operation would you use to determine how much farther a hike to the lake is than a hike to the mountain summit?

f. How much farther is a hike to the lake than a hike to the mountain summit?

2. Chloe is starting a painting business. The supplies she purchased for the business cost $\$825$. She will earn a fee of $\$80$ for each room that she paints. Chloe wants to determine how many rooms she will need to paint before she starts making a profit.

a. ○ Fees Chloe earns
○ Supplies Chloe buys
increase her profit.

b. ○ Fees Chloe earns
○ Supplies Chloe buys
decrease her profit.

c. Write a value in the blank for each absolute value and a word in the remaining blank to complete each sentence.

To find the revenue, Chloe multiplies the number of

rooms she paints by |_____|.

To find the profit, Chloe subtracts |_____| from the _______________.

d. In the table, complete the mathematical statement and then perform the operations to determine if the number of rooms painted would result in a profit.

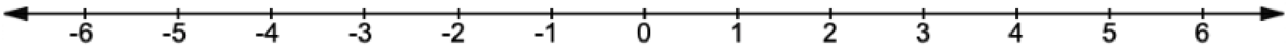
Number of Rooms	Mathematical Statement	Profit (Yes or No)
6	6|_____| — |_____|	
10	10|_____| — |_____|	
13	13|_____| — |_____|	
16	16|_____| — |_____|	

4.12.3 Try It: Operations with Absolute Value in the Real World

Logan, Mia, and Xavier all live on the same street as their school, which runs from east to west.

- Logan lives 4 blocks to the west of the school.
- Mia lives 5 blocks to the east of the school.
- Xavier lives 2 blocks to the west of the school.

1. Use the number line to model the position of each house and the school along the street. Label the location of each person's house and the school.



2. Work with your partner to write the mathematical sentence using absolute values. Then, use the mathematical sentence to determine each answer.

   a. How far does Logan live from Mia?

   b. How far does Mia live from Xavier?

   c. How far does Xavier live from Logan?

   d. Was the same operation used to find each distance?

4.12.4 Collaboration: Halfway Point

Mount Ojos del Salado is the highest mountain in Chile. Its peak is $6,893$ meters above sea level. The lowest point of the Atacama Trench, just off the coast of Peru and Chile, is $8,065$ meters below sea level.

Work with your partner to answer each question.

1. Determine how to label the number line so that you can approximate a location on the number line that represents Mount Ojos del Salado, Atacama Trench, and sea level.

2. Label the number line and place a point at the approximate location for Mount Ojos del Salado, Atacama Trench, and sea level.

3. Write a mathematical sentence using absolute value to find the difference in elevations between Mount Ojos del Salado and the Atacama Trench.

4. With your partner, discuss how the halfway point between the peak of Mount Ojos del Salado and the Atacama Trench can be found. Write down all of the strategies that were discussed.

5. Explain how to determine if the halfway point's elevation is above sea level or below sea level.

4.12.5 Your Turn!

1. The captain of a fishing vessel is standing on the deck at 23 feet above sea level. He holds a rope tied to a fishing net below him that is underwater at a depth of 38 feet.

   a. Draw a model of this situation using the number line.

   b. Write a mathematical sentence using absolute value to determine the length of the rope.

2. Aryan owes $\$325.46$ for recent purchases he made using his credit card, and he owes his parents another $\$187.27$.

   a. Write a mathematical statement using absolute value that can be used to determine Aryan's total debt.

   b. What is the value of Aryan's total debt?

3. When Brenda places a package of chicken in the freezer, the chicken is at a temperature of 41° Fahrenheit ($^{\circ}\mathrm{F}$). While in the freezer, the temperature of the chicken changes $^{-}2^{\circ}\mathrm{F}$ every hour until it reaches the temperature of the freezer. What is the temperature of the chicken 3 hours after Brenda puts it in the freezer?

4.12.6 Wrap-Up: Reflection

1. Check all of the statements below that describe your understanding of and ability to work with absolute values.

- ☐ I can compare the absolute value of two numbers.
- ☐ I need support on how to compare the absolute value of two numbers.
- ☐ I can perform operations that involve the absolute value of two numbers.
- ☐ I need support to perform operations that involve absolute value.
- ☐ I can interpret comparisons of absolute values in real-world contexts.
- ☐ I need support interpreting comparisons of absolute values in real-world contexts.
- ☐ I can perform operations that involve absolute value when solving real-world problems.
- ☐ I need support to perform operations that involve absolute value when solving real-world problems.